

To put a name together, you must identify these four parts:

Substituents	Parent Chain	Unsaturation	Principal Functional Group
Use prefix	Use prefix/root	Use suffix	Use suffix

Remember from class:

Substituents = Any group coming off of parent chain

Parent Chain = Longest continuous carbon chain that contains the most principal functional groups

- If no functional groups are present or doing this results in a tie, then the longest continuous carbon chain that contains the most double bonds
- If still resulting in tie, then longest continuous carbon chain that also contains the most triple bonds

Unsaturation = Presence of any double or triple bonds (or lack thereof)

Principal Functional Group = Highest priority functional group (refer to table provided in class for order)

Tip: Use this table to help organize your pieces!

Substituents	Parent Chain	Unsaturation	Principal Functional Group
<u>Use prefix</u>	<u>Use prefix/root</u>	<u>Use suffix</u>	<u>Use suffix</u>

To then actually name the molecule:

1. Identify principal functional group
 - a. Identify functional group and determine which one has the highest priority.
 - i. Choose appropriate suffix for this functional group and number based on Step 3 once that has been completed
2. Identify parent carbon chain
 - a. Find the longest continuous carbon chain that contains the most “principal functional groups” if any were identified
 - b. If no functional groups are present or 2a results in a tie, choose the chain that contains the most principal functional groups and the most double bonds.
 - c. If no functional groups are present and no double bonds are present or 2b. results in a tie, choose the chain that contains the most principal functional groups, the most double bonds, and the most triple bonds.
3. Number the parent carbon chain that gives the lowest possible number to the highest priority item
 - a. Principal functional group
 - b. Double bond
 - c. Triple bond
4. Identify unsaturation status
 - a. Are there double and/or triple bonds?
 - i. If yes, locate them and use the lowest number carbon the bond is involved with to number it
 - b. Use a counting prefix modifier (di, tri, tetra, etc.) to indicate how many there are if more than 1 exists
5. Locate substituents
 - a. Everything else that has yet to be taken into account is now considered a substituent of the parent chain.
 - i. Number them all based on the numbering done from Step 3
 - ii. If you have any substituents that are identical to each other, group them together and place the appropriate counting prefix modifier in front of it. Numbers associated with all of these substituents will be placed in front of the counting prefix.
6. Put all substituents in alphabetical order
 - a. “Iso-”, “cyclo-”, and “neo-” groups are counting in alphabetizing
 - i. Note, in this class, we only have seen “iso-” and “cyclo-”
 - b. “tert-”, “sec-”, “n-” prefixes are not counted in alphabetizing
 - i. Note, in this class, we have only seen “tert-”
 - c. Counting prefix modifiers are also not counted in alphabetizing
7. Combine the pieces
 - a. Use commas between numbers
 - b. Use hyphens anytime you change from numbers to letters